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A dual-component framework of interactive attention: Gaze markers of expertise and engagement in split-attention gameplay

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ABSTRACT

Video games require complex coordination of motor skills and active perception, yet the cognitive mechanisms underlying the transition from novice to expert remain under-examined. This study utilizes a custom-built 2D puzzle-platformer, *Vacillate*, to investigate how Game Literacy moderates the distribution of visual attention in a split-attention environment. Drawing on eye-tracking data from 52 participants, we identify a distinct state of foveal dependency in novice players, characterized by reactive avatar monitoring. Conversely, increased expertise predicts visual economy, where players reallocate gaze toward predictive goal information. Results reveal that Game Literacy is a robust negative predictor of avatar-dwell time ($R^2 = 0.33$), while self-reported Enjoyment relates to slower, more deliberate saccadic processing. We propose the Dual-Component Framework of Interactive Expertise, which dissociates mechanical proficiency (Game Literacy) from affective engagement (Enjoyment). This framework posits that digital expertise is defined not by raw performance speed, but by the strategic economy of visual resources, a fundamental shift from reactive foveal dependency to proactive environmental scanning.

1. Introduction

Interaction in video games requires a sophisticated coordination of motor skills, active perception, and real-time problem-solving, creating a unique set of cognitive demands that distinguish video games from passive media (Costykian, 2005; Wolf, 2007). Game designers typically focus on creating gameplay features that motivate users to engage and progress (Brathwaite & Schreiber, 2009, pp. 2–5), but there is an underlying challenge in game design: the management of the player's limited attentional and cognitive resources. Costykian (2005) highlights that game design is among the most difficult of design disciplines precisely because it requires anticipating, responding to, and guiding the unpredictable and diverse ways players interact with the game. For the player, mastery is not just a matter of motivation or intent but requires attentional efficiency. However, while designers can track outcome-based metrics like score or completion time, the internal attentional and cognitive bottlenecks that prevent a player from achieving mastery remain largely invisible. It remains unclear how a player's limited attention is allocated between the mechanical requirements of avatar control and the cognitive demands of strategic planning; this study investigates this split-attention conflict.

Whether navigating the high-stakes scanning of first-person shooters or the spatial logic of puzzle-platformers like *Little Nightmare* (Tarsier Studios, 2017), players must balance competing demands of visual monitoring and real-time problem-solving. This ability to efficiently allocate attention matures with experience, defining the transition from a novice's reactive visual style to an expert's proactive strategy. This study utilizes eye-tracking metrics to map this transition, investigating the specific attentional costs incurred as players move toward the strategic engagement characteristic of expertise. While current design frameworks can describe what a player does, they largely fail to explain the cognitive cost of how they achieve it. The significance of this study lies in its ability to quantify the ergonomic boundaries of human-computer interaction in multi-tasking environments. By identifying

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the visual markers that distinguish novice from expert strategy, we take a necessary step toward understanding how users navigate and control spatially separated data streams. While the mechanisms of visual economy explored here may eventually inform the design of adaptive interfaces or digital training tools, the present study focuses specifically on establishing the foundational oculomotor signatures of the split-attention conflict within a controlled, interactive gameplay context.

1.1. A dual-component framework of interactive attention

To examine these visual markers, this study proposes a dual-component framework that distinguishes between the structural allocation of gaze and the qualitative nature of visual engagement. See Fig. 1 for a diagram of the Dual-Component Framework.

The first component, attentional efficiency, is expertise-driven and concerns the visual economy required to automate motor feedback loops (i.e., reducing the tax on attention caused by avatar monitoring). The second component, attentional state, is affect-driven and concerns how a player's internal state, such as enjoyment or flow, is revealed by their scanning patterns. By establishing these two distinct dimensions, we provide a theoretical lens through which to evaluate how Game Literacy (as a measure of expertise) and subjective experience (as a measure of engagement) together shape the player's ability to navigate and manage split-attention conflicts.

In this framework, we distinguish between theoretical constructs and their operationalized indicators. We define attentional efficiency as the overarching goal of visual resource management, which is observed through a player's visual economy, i.e. the strategic distribution of gaze. Similarly, while we seek to understand the player's attentional state (broadly conceptualized as Flow; Csikszentmihalyi, 1990), we specifically measure self-reported enjoyment as the primary indicator of this state. While enjoyment is a core component of the Flow experience, we acknowledge that Flow represents a more complex, multi-dimensional state of deep absorption. Finally, we distinguish between expertise as a broad theoretical concept and Game Literacy as our specific, measurable metric of a player's familiarity with game conventions and mechanics.

1.2. Attentional Efficiency

The development of expertise in highly dynamic digital platforms, such as video games, is partly driven by a highly developed interactive visual economy (Hadnett-Hunter et al., 2019; Juvrud et al., 2024). While the skilled player quickly directs their gaze toward critical information for both navigation and long-term planning, such as attention to incoming targets well ahead of their collision with the player character (Juvrud et al., 2024), the novice often struggles to automate these processes. The study of expertise in complex digital environments is fundamentally a study of attentional conflict; specifically, how a player manages the split-attention demand between necessary visuomotor control and information extraction in real-time.

While performance metrics, such as speed or score, provide clear outcome data, they often obscure the underlying attentional and visuomotor processes that define skill acquisition. The development of expert-level play requires more than simply mastering controls (Brown et al., 2009); it necessitates an optimized allocation of visual resources, allowing the player to selectively identify and process relevant information within the game's streams of data. Players must continuously decide where to look, what to prioritize, and how to integrate that information with ongoing motor actions.

Modern video games, particularly those that blend simultaneous motor and cognitive challenges, impose a fundamental split-attention conflict on the player. Puzzle-platformers, for example, combine real-time navigation with visual scanning, pattern recognition, and strategic planning. An example of this conflict is found in the split-screen cooperative mechanics of *It Takes Two* (Hazelight Studios, 2021). In this context, a player must often perform precise platforming on their side of the screen while simultaneously extracting critical puzzle information from their partner's spatially separated half of the screen. This conflict forces a continuous, resource-intensive shift in gaze and cognitive load between distinct, critical areas, such as navigating a persistent player character and solving a puzzle in a different location.

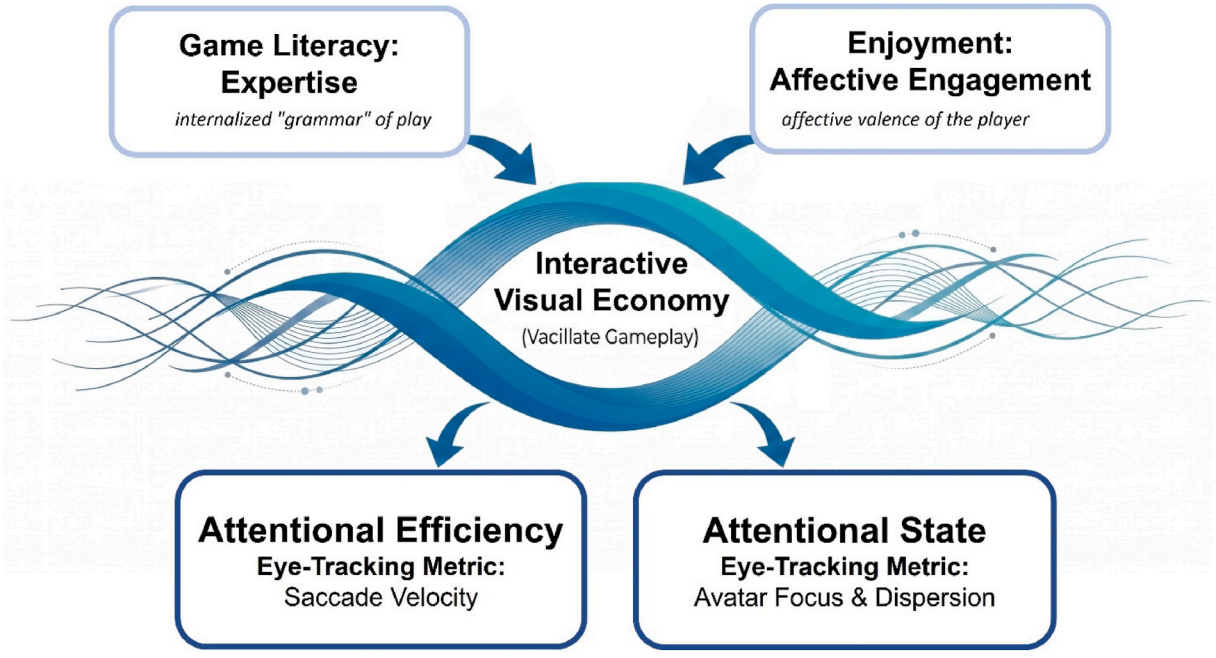


Fig. 1. The Dual-Component Framework of Interactive Visual Strategy
Note. This figure illustrates the theoretical relationship between the cognitive component (game literacy) and the affective component (enjoyment) as predictors of player visual behavior. Game literacy is hypothesized to drive motor automaticity and reduce foveal dependency, while enjoyment is proposed to moderate the transition into a positive attentional state. Together, these components define the dual-framework approach to analyzing interactive visual economy.

1.3. Game literacy

To evaluate the proposed two-component framework, it is first necessary to refine how we define and measure player expertise. While traditional metrics often rely on raw playtime metrics (hours, years, etc.), total duration may not fully capture the nuanced individual differences in how players process interactive systems across play. Instead, we propose that the "attentional efficiency" defining an expert is a manifestation of Game Literacy (Gee, 2003).

While literacy was traditionally defined by linguistic decoding, contemporary frameworks emphasize emergent literacies that account for digital and procedural fluency. In this study, we define Game Literacy not just as technical proficiency, but as the internalized understanding of a game's "grammar": its rules, mechanics, and spatial conventions (Gee, 2003; Juul, 2005).

Consider the spatial mapping of a standard game controller. To a player with low experience, the array of joysticks, triggers, and buttons represents a significant cognitive barrier, often requiring the player to physically look away from the screen to verify their hand positioning. This is a hallmark of the cognitive stage of motor learning (Fitts & Posner, 1967). Through repeated exposure, however, these physical mappings are internalized and become automatic (proprioceptive automaticity). This allows experienced players to offload the mechanics of input to the motor cortex, freeing foveal attention to focus exclusively on task-critical screen information rather than the interface itself (Kirsh & Maglio, 1994). Experienced players are now no longer learning a new language but rather adapting a known mental schema to a new hardware dialect. This literacy is both ingrained and highly transferable. Because the fundamental grammar of the controller is already automated, the player can maintain their visual focus on the screen, thereby bypassing the manual attentional tax that would otherwise hinder their performance.

Similarly, familiarity with common design patterns and gameplay conventions (or genre grammars) allows experienced players to navigate and learn new games more quickly. By utilizing internalized common design patterns (e.g., spikes cause damage) and genre conventions (e.g., you can jump over spikes in platformers), experienced players can anticipate the functionality of various systems and mechanics without explicit instruction (Björk & Holopainen, 2005). This predictive ability transforms the gameplay experience from a reactive search for meaning into a proactive application of known strategies. Much like an experienced reader uses context clues to skip over familiar words, a game-literate player uses their knowledge of genre conventions to bypass the "search-heavy" phase of learning and move directly into strategic engagement.

1.4. The "attentional tax" on players

To fully understand the mechanisms that underlie a player's visual economy, or the tension between immediate visuomotor execution and long-term information extraction, we posit that player engagement is built around two core, related concepts: attentional efficiency and attentional state. The first concept concerns efficiency and is hypothesized to be a direct consequence of expertise. Specifically, we predict that less experienced players will pay a measurable "attentional tax." While we colloquially use this term to describe the temporary reallocation of visual resources toward motor-control feedback loops, such as focusing on the avatar or "avatar monitoring", this study specifically investigates whether this "tax" is a functional requirement for early-stage motor learning. By examining to what extent novices rely on the foveal monitoring of their avatar, we aim to determine if this behavior represents an inefficient drain on resources, or a necessary foveal dependency. Here, such a foveal dependency is hypothesized to be a bridge for the motor system that has yet to achieve proprioceptive automaticity. Framing the tax in this way allows us to investigate not just its presence, but its underlying cognitive role within the transition from novice to expert.

Consider *Super Mario Bros.* (Nintendo, 1985), a side-scrolling platformer that requires precise jumps and forward momentum. We hypothesize that a skilled player can anticipate a jump while maintaining forward momentum—indicated by predictive shifts in attention to incoming environmental changes. New players, by contrast, will struggle with momentum and instead focus their attention on the avatar as they learn precision.

The second concept concerns the player's affective state. We suspect that a highly developed interactive visual economy is not merely a marker of expertise, but the primary facilitator of positive attentional states such as flow (Csikszentmihalyi, 1990) and enjoyment. This may manifest in qualitatively different scanning behaviors during play, e.g., controlled versus erratic saccades. If we apply this to *Super Mario Bros.,* we hypothesize that a highly skilled player will achieve a flow state and be able to adjust their predicative movements to match the increasing challenge and momentum of the game (e.g., increased demand for precision jumps).

1.5. Eye-tracking metrics

The significance of a player's gaze, or visual attention, while playing a game lies in its role as a real-time proxy for cognitive prioritization (Meerhoff et al., 2018). In these high-stakes environments, gaze is not only a record of visual focus, but also an empirical metric of what the player values at any given second. By quantifying these shifts in attention, we gain an objective window into the player's moment-to-moment strategies and the mechanisms of skill acquisition.

Eye-tracking captures where, when and how users allocate their attention on the screen, providing continuous, high-resolution data on visual attention, perception and cognitive load (Jacob & Karn, 2003; (Juvrud et al., 2024); Poole & Ball, 2006). Because eye movements are closely linked to cognitive processes (Mahanama et al., 2022), analyzing gaze behaviour offers insight into how users perceive, process, and interact with information in real time. In human-computer interaction research, eye-tracking is widely used because it provides objective evidence of how users navigate and process interactive systems. By linking gaze data to task performance, it becomes possible to assess attention allocation, cognitive load and usability in ways that self-report methods cannot capture (Jacob & Karn, 2003; Poole & Ball, 2006).

To empirically test the dual-component framework, we selected specific oculomotor metrics that serve as proxies for our theoretical constructs. For attentional efficiency, we selected dwell time, fixation count, and revisit count within the avatar area of interest (AOI) as the primary indicators of foveal dependency. Dwell time (Joseph and Murugesh, 2020) provides a cumulative measure of visual attention, representing the total cost paid to motor monitoring. High fixation and revisit counts specifically index a reactive feedback loop, where the player must constantly return their gaze to the avatar to verify its position (Jacob & Karn, 2003). These metrics were chosen over time-to-first-fixation and other related metrics because the attentional tax is a persistent burden throughout gameplay, rather than a single event at the start of a trial.

For attentional state, we selected peak saccade velocity within the puzzle AOI as a proxy for cognitive fluency. Unlike fixation duration, which often reflects local processing difficulty, saccade velocity captures the quality of the search itself. Based on prior research (Skulmowski & Rey, 2017), high-velocity saccades are indicative of frantic search patterns under high cognitive load, whereas lower velocities reflect the deliberate, structured scanning associated with enjoyment and Flow. Finally, for global visual economy, we used fixation dispersion (Juvrud et al., 2024; Wang et al., 2024) to index strategic focus. Broad dispersion suggests an exploratory, search-heavy phase where the player is uncertain of what information is relevant. Conversely, a compressed dispersion reflects an expert's ability to selectively prioritize task-relevant regions, filtering out environmental noise.

Beyond specific areas of interest, the global spatial distribution of a player's gaze provides a holistic marker of their internal mental model. In complex digital environments, expertise is often characterized by visual chunking, or the ability to group related information and ignore redundant environmental noise (Haider & Frensch, 1999). As a result, experts may exhibit lower spatial fixation dispersion, as their gaze is pulled toward highly useful clusters of information. Novices, however, who lack a more developed schema of the game's logic, may engage in more diffuse, exploratory scanning patterns. In this way, global dispersion serves as an index of attentional uncertainty. While a more scattered gaze reflects a reactive search for relevant cues, a compressed gaze reflects a proactive, goal-oriented strategy where the player knows exactly where the critical information will appear (Van Gog and Scheiter, 2010).

1.6. Identifying the gaps

Beyond traditional interfaces, games represent ideal contexts for studying how individuals coordinate visual attention under high cognitive load. Players must simultaneously monitor avatars, environmental cues, and UI elements, making rapid decisions that link perception to motor output (Iqbal, 2015). A consistent distinction in game studies emerges between expert and novice gaze behavior: experienced players typically display narrower gaze dispersion, fewer unnecessary revisits, and more focused attention on task-relevant areas (Juvrud et al., 2024; Wang, Yang, Hu, Tang, Yu, 2024). Furthermore, gaze metrics serve as subtle indicators of affective engagement. Slower, more controlled saccades and structured scanning are generally associated with smoother gameplay and higher enjoyment, whereas erratic, high-velocity movements indicate cognitive overload or stress (Joseph & Murugesh, 2020).

Despite these insights, a central challenge remains that most commercial games confound motor, perceptual, and cognitive demands. In a commercial title such as *Super Mario Bros.*, it is difficult to isolate whether a player's gaze shift is a response to a mechanical requirement (e.g., jumping) or a cognitive one (e.g., puzzle solving). While research suggests that expertise and affective state interact to shape visual strategy (Latham et al., 2013), this relationship has rarely been explicitly examined in a controlled environment that isolates these variables.

Specifically, it remains unclear whether high levels of player enjoyment are a causal consequence of achieved visual efficiency or if these states develop independently. Furthermore, current research has yet to determine if an economic allocation of attention can manifest during complex tasks before motor skills (e.g., avatar control and navigation) are fully automated. Finally, there is a lack of clarity regarding how these oculomotor processes unfold when players are forced to reconcile the tension between immediate visuomotor execution and the extraction of spatially separated information.

To address this gap, we developed *Vacillate*, a custom-built 2D puzzle-platformer designed to induce a structured split-attention conflict. By presenting two spatially separated but interdependent tasks, real-time platforming and puzzle-solving, the game creates distinct, non-overlapping areas of interest. This design allows for a granular analysis of gaze allocation as players decide where to prioritize their attention, shifting between these areas of interest. *Vacillate* enables us to investigate how expertise (operationalized as Game Literacy) and affective state (captured through enjoyment) jointly shape the visual economy required to resolve internal attentional conflicts.

1.7. Current study

By integrating oculomotor dynamics (fixation patterns and saccade dynamics) with individual differences in expertise (such as prior expertise in video games), this research aims to establish a two-component framework that links player experience (attentional efficiency) to affective engagement (attentional state). To empirically test this framework without the confounding effects of game familiarity or prior game-specific knowledge, we developed a custom platformer that serves as a controlled environment specifically designed to isolate shifts in attention. This was accomplished through the creation of distinct, spatially separated zones of predictive information, thereby allowing us to map the precise moment when a player moves from the taxed state of a novice to the strategic state of an expert.

Whether navigating the high-stakes scanning of first-person shooters or the spatial logic of puzzle-platformers like *Little Nightmare* (Tarsier Studios, 2017), players must balance competing demands of visual monitoring and real-time problem-solving. This ability to efficiently allocate attention matures with experience, defining the transition from a novice's reactive visual style to an expert's proactive strategy. This study utilizes eye-tracking metrics to map this transition, investigating the specific attentional costs incurred as players move toward the strategic engagement characteristic of expertise. While current design frameworks can describe what a player does, they largely fail to explain the cognitive cost of how they achieve it. The significance of this study lies in its ability to quantify the ergonomic boundaries of human-computer interaction in multi-tasking environments. By identifying the visual markers that distinguish novice from expert strategy, we take a necessary step toward understanding how users navigate and control spatially separated data streams. While the mechanisms of visual economy explored here may eventually inform the design of adaptive interfaces or digital training tools, the present study focuses specifically on establishing the foundational oculomotor signatures of the split-attention conflict within a controlled, interactive gameplay context.

Beyond theoretical contributions, this work has applied implications for game and human computer interaction (HCI) design. If specific gaze patterns reliably indicate overload, uncertainty, or inefficient visual strategies, then these patterns could inform adaptive systems that scaffold novices toward more efficient engagement or inspire tutorial designs that reduce the attentional tax on new players (Paas et al., 2003; Sweller, 1994). This could be particularly relevant for video games that are explicitly designed for training, such as simulations. By integrating these considerations into a controlled experimental context, this study situates itself at the intersection of cognitive science, HCI, and player-centered design.The present study investigates how player experience and affective engagement shape visual strategy in a split-attention environment. We move beyond traditional performance outcomes by utilizing eye-tracking to reveal the underlying cognitive processes of gameplay. Based on the proposed two-component framework of attentional efficiency and attentional state, we test the following formal hypotheses:***Component 1: Attentional Efficiency***.

We hypothesize that higher game literacy facilitates the automation of motor control, reducing the need for foveal feedback from the player character. Consequently, we expect expertise, as measured by game literacy, to be a primary predictor of reduced avatar monitoring behavior.

- H1a: Game literacy will negatively predict the percentage of dwell time within the avatar AOI.
- H1b: Game literacy will negatively predict the fixation count within the avatar AOI.
- H1c: Game literacy will negatively predict the revisit count within the avatar AOI.

1.7.1. Component 2: attentional state

We hypothesize that positive affective states, such as enjoyment and flow, are characterized by more deliberate and less erratic visual scanning. In contrast, cognitive friction or frantic searching will manifest as high-velocity saccadic movements.

- H2: Self-reported enjoyment will negatively predict peak saccade velocity within the puzzle-solving AOI.

1.7.2. Global visual economy

Finally, we examine the global spatial structure of attention as a marker of refined strategic expectations. Broad, diffuse scanning suggests uncertainty, while compressed gaze clusters reflect information prioritization.

- H3: Game literacy will negatively predict global fixation dispersion across the entire play area.

To evaluate these hypotheses, we employed an experimental design utilizing a custom-built 2D puzzle-platformer, *Vacillate*, designed to isolate the split-attention conflict between motor control and cognitive processing. Gaze behavior was recorded from 52 participants using high-resolution eye-tracking. We then used a series of multiple linear regression models to analyze the predictive power of game literacy and enjoyment on oculomotor metrics across three distinct areas of interest. This approach allows for a granular assessment of how individual differences in expertise and engagement shape the visual economy of gameplay.

In summary, this study offers three primary contributions to the field of interactive attention. First, we introduce *Vacillate*, an experimental paradigm designed to isolate the fundamental cognitive conflict of split-attention within a controlled eye-tracking environment, avoiding the confounding variables inherent in commercial titles. Second, by simultaneously modeling game literacy and affective engagement as predictors, we provide empirical evidence for potentially differing dynamics in gaze behavior, disentagling the cognitive requirements of control and the affective rewards of engagement. Finally, we propose a dual-component framework that offers a new theoretical lens for understanding the transition from foveal dependency to attentional agency.

Critically, this study is among the first to jointly model expertise and affective engagement as simultaneous predictors of dissociable gaze signatures, representing a methodological advance over prior research that has largely examined these variables in isolation.Together, these contributions provide a mechanistically grounded account of how expertise and affect interact to shape the visual grammar of digital play (see Table 1 for a glossary of key constructs). By integrating oculomotor measurement with game literacy theory, this study provides a genuinely cross-disciplinary perspective that bridges the gap between HCI, cognitive science, and educational psychology.

2. Method

2.1. Participants

A convenience sample of 52 participants was recruited from the DIS Stockholm campus and housing buildings. This recruitment site accessed a population of international university students with diverse technological backgrounds. Seven participants were excluded prior to the final analysis. Specifically, three participants were excluded because of poor eye-tracking calibration quality, defined as a gaze accuracy error exceeding 0.5° of visual angle, which rendered their spatial data unreliable. Four further participants were excluded due to technical errors in the game (game bugs and crashing). The final analysis is based on 45 participants who provided complete eye-tracking data. The study maintained a 98.2% completion rate of the game, indicating that the experimental task was achievable even for those with minimal prior gaming experience. The sample (33 Female, 14 Male, 3 Nonbinary, 1 Male/Apathetic Gender, 1 Queer) had a mean age of 24.96 years (SD = 7.27, range 17–51). To ensure a robust test of the expertise-driven attentional tax, the recruitment successfully targeted a novice-heavy demographic: 67.3% of the sample reported playing video games for 2 h or fewer per week. All participants had no prior experience with the study's specific game environment (*Vacillate*). This distribution allowed for a clear contrast between true novices and those with higher baseline Game Literacy. All participants provided informed consent and were compensated with a €5 gift card. The methods and protocols of the study were conducted in accordance with the standards specified in the 1964 Declaration of Helsinki and approved by the local ethics committee, Informed consent was obtained from all individual participants included in the study. This study was reviewed and approved by the Swedish Ethics Committee (Dnr 2026-01611-01). All procedures performed in studies involving human participants were in accordance with the ethical standards of the national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

2.2. Materials

2.2.1. Vacillate video game

The custom developed game, *Vacillate,* was a 2D side-scrolling platformer in which users, playing as a small bird, traversed two parallel worlds (see Fig. 2). Visually and thematically contrasting, both environments were simultaneously visible on screen during gameplay and were mirrored across the ground plane. During the introductory cut scenes, the participant was introduced to the player character (avatar), the narrative, and the game's basic objective. Participants used a gamepad to play and were familiarized with the control scheme during a short tutorial level. The *Vacillate* game can be downloaded and played at https://katesalembier.itch.io/vacillate.

Unlike studies utilizing commercial games, which often contain confounding variables such as dynamic camera angles or complex HUDs, the custom-built paradigm *Vacillate* was designed to isolate the core cognitive conflict of split-attention. The fixed camera allows for a consistent avatar AOI, while the two distinct yet simultaneously visible environments created an ecologically valid but experimentally isolable attentional conflict. By providing the game as an open-source tool for replication, we offer a controlled experimental affordance for studying the cognitive architecture of interactive play that commercial titles cannot provide.

The core mechanic of *Vacillate* centered on traversing platforms, dodging obstacles, and solving puzzles. Players interacted with objects in order to collect “keys” required to unlock “gates.” There were four keys and four corresponding gates spread across both environments. This required players to frequently rotate between environments, an action that could only occur at designated interactable points of the map, to complete the game. Mirrored across the two environments, each

Table 1
Key constructs of player visual engagement.

Term	Operational Definition	Theoretical Role
Game Literacy	A player's internalized understanding of game “grammar,” rules, and spatial conventions.	Primary predictor of Attentional Efficiency.
Enjoyment	The degree of affective interest and positive engagement with the game task.	Predictive of retention, but distinct from mechanical expertise.
Foveal Dependency	The tendency to anchor gaze to the player avatar for continuous visual feedback.	An oculomotor manifestation of the Attentional Tax faced by novices.
Visual Economy	A state of gaze where visual attention is allocated only to high-value, task-relevant information.	The primary behavioral outcome of high Game Literacy.
Peak Velocity	The maximum speed reached during a saccadic eye movement.	An objective index of Motor Automaticity and reduced cognitive load.

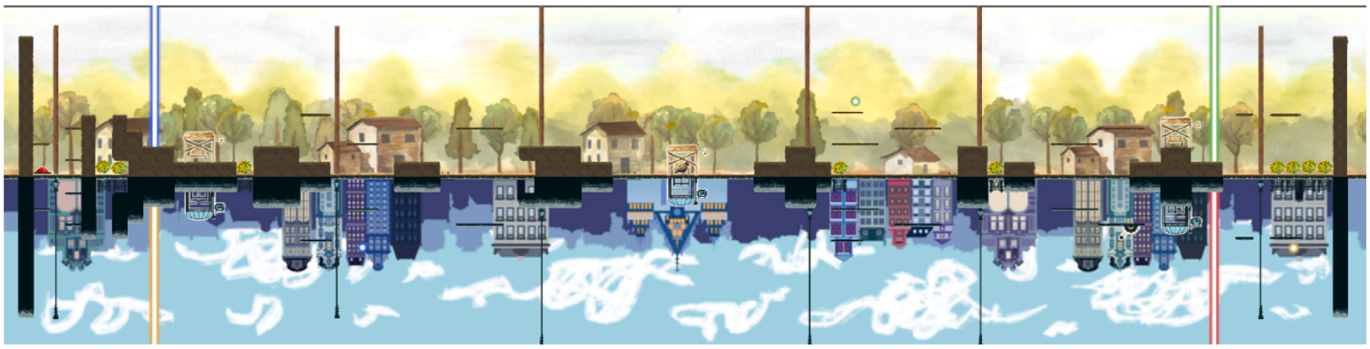


Fig. 2. Overview of the vacillate game level.

key contained two correlated counterparts: the collectible key object, which was invisible but interactable, and a "hint" which was a visual representation of the key's location that was not collectible (Figs. 3 and 4). Because all other game elements (platforms, obstacles, interactable objects, etc.), were symmetrically reflected across the ground plane, players were required to observe the environments and note differences between the two environments to solve puzzles and progress. This puzzle mechanic provided the main gaze behavior (i.e., attention across the two environments) and thus defined the distinct areas of interest (see Fig. 5).

Players began in the center of the map, and obstacle collisions forced players to return to the center of whichever environment they were in when they encountered the obstacle. There was no other punishment for obstacle collisions and no limit on attempts (i.e., lives). The game was designed to be non-linear, so there was flexibility in which order the puzzles could be completed. However, some interactable objects necessary to solve puzzles could only be found behind locked gates, so not every combination was possible. This required players to develop a multi-step strategy and use problem-solving to determine a successful sequence of events. Importantly, it was not possible to progress through the puzzles without using visual information from the opposite environment.

Level Structure and Objectives. The level requires players to collect four color-coordinated keys (Green, Blue, Yellow, and Red) to open their corresponding gates. While players begin in the center of the level and can approach the first three puzzles in any order, the Red gate serves as the final bottleneck and can only be cleared once the other three are complete. Game completion is defined by opening the Red

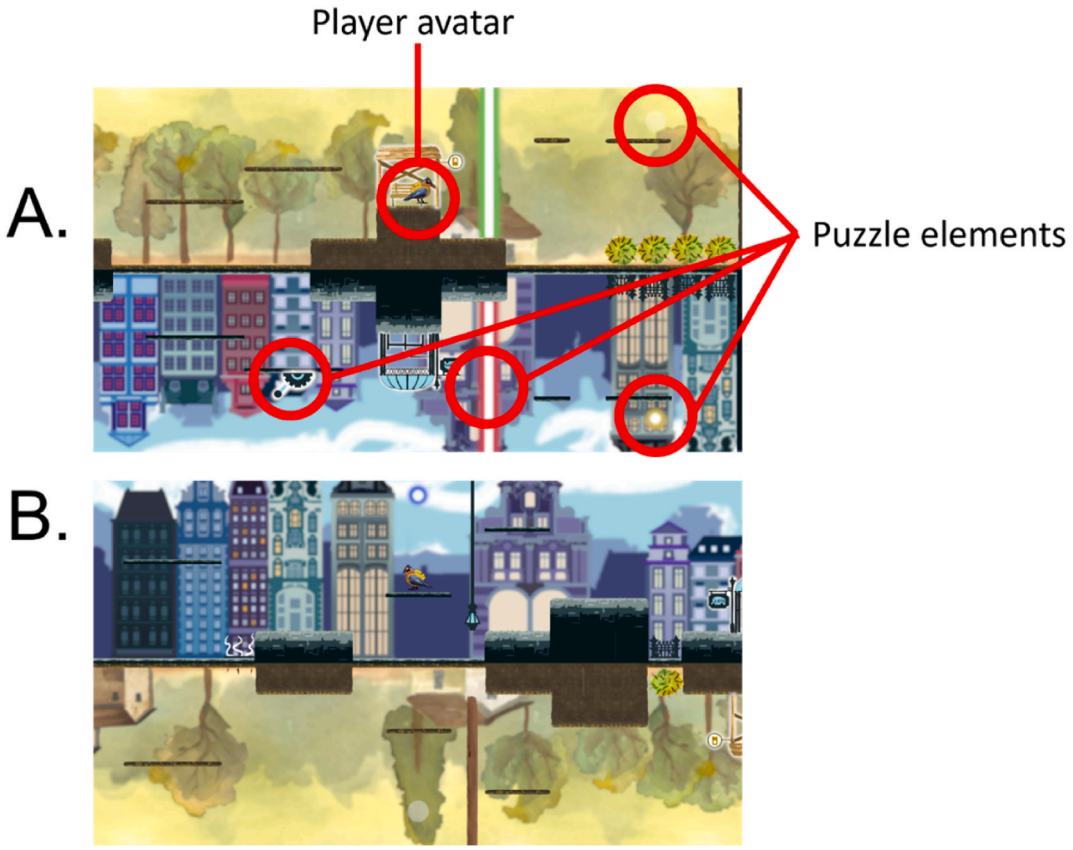


Fig. 3. Overview of Level Information, Layout, and the Flip Mechanic
Note. Screenshots of the game Vacillate, (A) illustrating the mirrored-environment design with puzzle elements (keys, gates, obstacles) on the bottom used to solve elements in the top player area. At certain points in the level, the player can 'flip' the world (B), which was necessary to solve some puzzles in addition to using information from the bottom environment.

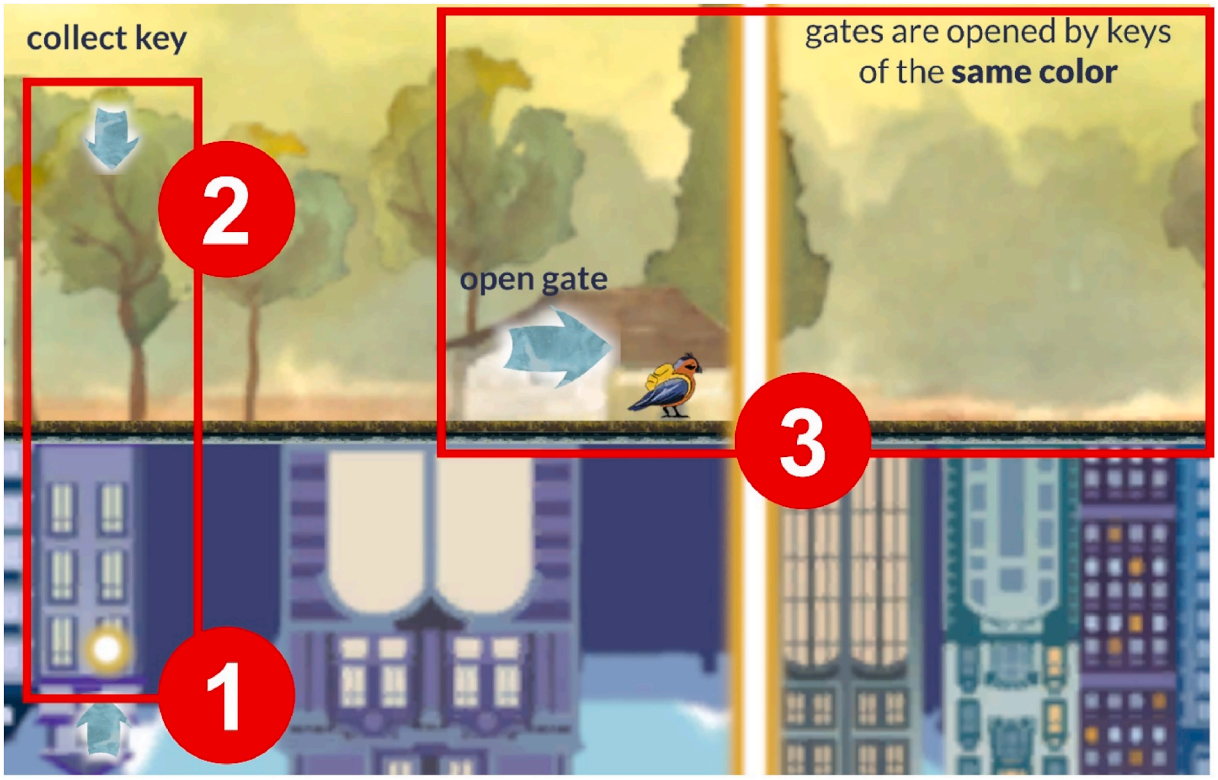


Fig. 4. Example Puzzle Solving Sequence
Note. A screenshot of the turorial section of the game demonstrating the order of actions to solve one puzzle type. (1) The player needs to identify the hidden information in the lower screen (a light indicator), then proceed to (2) go to that mirrored location in the top screen to obtain a key, so that the play can (3) open a gate that matcher the color of the key somewhere in the level. Note: text in the figure is the same text that appears to players during the tutorial.



Fig. 5. Visualization of the Four AOIs Used for Data Extraction
Note. AOIs were defined for data extraction: the player avatar area (red), the puzzle area (blue), fixed on the player avatar (green), and global visual ergonomics area (the entire screen; yellow).

gate, reaching a visually distinct final "world flip point," and triggering a concluding cutscene.

World-Flip Mechanic. Players transition between the two sides of the screen (City or Village) using "World-Flips." These flips are not instantaneous or universal; they can only be initiated at six specific locations (three mirrored pairs of bus stops). To trigger a flip, the player must stand within the bus stop AOI and press the interact button (E on keyboard; Left/West button on a gamepad). The avatar stays in place, but the entire game world flips so that the player character is at the equivalent coordinates in the opposite environment, allowing them to progress through the mirrored landscape.

Puzzle Mechanics. Each puzzle is built on a cross-screen hint logic:

the visual hint for a key's location is always located on the opposite side of the screen from the key itself. The four puzzles increase in cognitive and mechanical complexity.

- Green Puzzle: A basic location-matching task where the player identifies the hint on one side and navigates to the equivalent position on the other.
- Blue Puzzle: A timing-based task where the key is moving; players must monitor the hint moving on the opposite side to time a double-jump correctly.
- Yellow Puzzle: A multi-step task requiring the player to flip a yellow lever to reveal the hint and activate a moving platform necessary to reach the key.
- Red Puzzle: A memory-based task. Players must stand on a red button to reveal the hint; however, the hint disappears as soon as the player leaves the button, requiring them to hold the spatial coordinates in memory while navigating to the opposite side of the screen.

Hazards and Obstacles. The level contains three functional types of hazards, each mirrored across the two worlds with distinct visual skins to match the theme (6 visual variants total).

- Stationary Hazards: Fixed spiky bushes (Village) or spiky gates (City) that remain in a constant position to obstruct movement.
- Falling Hazards: Logs (Village) or streetlights (City) that drop rapidly from the top of the screen before ascending slowly. These operate on a fixed, predictable schedule.
- Pop-up Hazards: Growing vines (Village) or steam vents (City) that extend from the ground on a repeating cycle. A small visual indicator remains visible even when the hazard is retracted.

Game Completion Operationalization. Total game completion time was selected as the primary performance metric to capture the holistic integration of manual navigation and cognitive puzzle-solving strategy. While metrics such as 'death count' primarily index mechanical error, completion time serves as a broader indicator of a player's efficiency in navigating the split-attention conflict. This choice aligns with our objective to evaluate how Game Literacy facilitates the transition from reactive survival to strategic environmental engagement.

Game completion time was operationalized as the total duration required for a participant to successfully solve the final puzzle in the sequence. While the experimental protocol established a 15-min maximum gameplay window, 100% of the retained sample successfully completed all game puzzles within this timeframe. Observed completion times ranged from 3.72 min to 11.79 min, with a mean duration of 8.83 min ($SD = 2.19$). Because no participant reached the 15-min limit, the completion time variable reflects the true, uncensored duration of successful task performance for every subject in the modeling.

Development and Pilot Testing. The design of *Vacillate* was refined through an iterative playtesting phase involving 34 members of the general public. This phase focused specifically on balancing the difficulty curve and ensuring the accessibility of the mirrored-environment mechanic. Feedback from this cohort led to adjustments in character movement physics and visual signaling for puzzle elements. This proactive validation, combined with a minimalist two-button control scheme, was intended to minimize mechanical frustration and ensure that the cognitive load measured was a product of the split-attention task itself rather than a poorly calibrated interface.

2.2.2. Eye-tracking

Hardware. Gaze data was collected using a screen-based Tobii Pro Fusion eye tracker. The Tobii Pro Fusion was operated at a 250 Hz sampling rate, which was selected to provide sufficient temporal resolution for capturing peak saccadic velocity, a key metric in our analysis of attentional state. The eye-tracker was mounted on a laptop with a 37 cm screen and 1920 x 1800 resolution on a monitor with a 60 cm screen. Participants were positioned in the center of the screen at an average distance of 60 cm. The software *iMotions* was used to calibrate the eye tracker using a standard 9-point calibration and record the gaze data (Gredebäck & von Hofsten, 2009).

Areas of Interest (AOI). AOIs were used to extract the relevant eye-tracking metrics. To extract the saccade and visual ergonomic data, we used the entire screen. Two additional AOIs divided the screen area in half to extract fixation metrics, with the top section for the avatar area and the bottom for the puzzle area. Finally, an AOI was created for the player character, fixed to the avatar model and accounting for vertical movement (jumps). Gaze data were mapped onto the four Areas of Interest (AOIs) with the following pixel dimensions: a global Screen AOI (1914 x 1082 px), a Top/Avatar Area (1912.2 x 547.8 px), a Bottom/Puzzle Area (1912.2 x 547.8 px), and a dynamic, character-fixed Avatar AOI (219 x 96 px).

The game was programmed using a fixed-center camera, meaning the avatar model was always centered on the screen with locked x,y coordinates and zero camera lead, allowing for consistent AOI tracking of the avatar. The decision to use a fixed-center camera was a deliberate methodological choice to isolate gaze behavior from camera-lead noise. By locking the player character's screen coordinates, we ensured that changes in the Avatar AOI metrics reflected the player's internal attentional shifts rather than reactive tracking of a moving camera target.

Fixation Count and Fixation Duration. From the eye-tracking data, we exported the number and duration of fixations as an indicator of cognitive load. This was based on prior studies that have demonstrated that the number and duration of fixations are positively correlated with cognitive load (Hessels et al., 2024), and in studies examining video games (Skaramagkas, 2023; Castaneda et al., 2016). Fixation duration has also been used to distinguish novices from experts in other domains related to schema acquisition (Dalveren & Cagiltay, 2025; Sheridan & Reingold, 2017).

Fixation detection was performed using the Tobii I-VT (Velocity-Threshold Identification) filter within the iMotions platform. The filter utilized a 20 ms window length and a velocity threshold of 30°/s to differentiate between fixations and saccadic movements. A minimum fixation duration of 60 ms was applied to the data. This duration threshold was specifically selected to account for the rapid informational intake required in a dynamic gaming environment. While higher thresholds (e.g., 100–200 ms) are standard in reading research, the 60 ms limit ensures that brief but cognitively significant glances toward moving interface elements or peripheral checks are preserved in the analysis, reflecting the high-speed visual processing typical of interactive tasks.

Saccade Count and Saccadic Velocity. From the eye-tracking data, we exported the number and velocity of saccades, the rapid movements of the eye between fixations. Prior studies have shown increased average peak saccadic velocity to predict an increase in cognitive workload (Bodala et al., 2014) and, more specifically, it has been shown to be more sensitive to germane load, the portion of cognitive effort allocated to meaningful learning activities such as building mental schemas and integrating new knowledge (Zu et al., 2018). Saccades can also be complementary to fixations in indicating skill level. Past studies with surgeons and map users show that in environments with a high cognitive load, novices made more saccadic movements compared to experts (Dalveren & Cagiltay, 2025; Keskin et al., 2020).

Downward Saccade Component. Saccade direction was linearized into a "Downward Saccade Component" using a sine transformation of the average angle. While this method collapses horizontal variance (effectively treating 30° and 150° as identical vertical vectors), it was chosen to specifically isolate the vertical transitions between the top-aligned avatar and bottom-aligned puzzle area. This allowed the directional component to be analyzed as a continuous linear variable within the hierarchical regression framework, directly reflecting the primary spatial dichotomy of the game's interface.

2.2.3. Self-report measures

Game Literacy Scale. The Game Literacy Scale assessed participants' prior gaming experience and familiarity with video games. The current scale was adapted by Rosenberg's (2011) Game Literacy Scale (see Supplementary Materials). All questions were reported on a Likert scale (*1 = Strongly disagree; 7 = Strongly agree*). An example item from the System Skills dimension is: "*I consider that I know how to do what I want when playing video games*." Additional added question assessed participants' time spent playing video games (*hours* per *typical week*).

The Game Literacy composite score was calculated as an unweighted sum of the Likert-scale items. Although the pre-survey included an item assessing participants' weekly hours of gameplay, this metric was utilized solely to provide a descriptive overview of the sample's general experience with digital games and was excluded from all subsequent predictive analyses to prioritize perceived expertise over raw time-on-task.

The instrument demonstrated robust internal reliability, yielding a total Cronbach's alpha of 0.93. These findings suggest the scale functioned as a reliable measurement instrument for this population.

Post-Game Enjoyment Question. The post-game enjoyment measure consisted of a single item administered immediately following the task. Participants were asked "*How much did you enjoy playing* Vacillate?" and provided responses on a 1–7 Likert scale (*1 = Did not enjoy at all; 7 = Enjoyed very much*). This item was treated independently from the literacy metrics in the subsequent saccade velocity analysis. A single-item measure was selected to assess global enjoyment to minimize a rebound effect on the immediately preceding oculomotor experience (see Bergkvist & Rossiter, 2007 for single-item validity in HCI).

2.3. Procedure

Participants were seated in an enclosed soundproof booth to provide informed consent and complete a pre-task survey. Following a 9-point eye-tracking calibration, they viewed a brief introductory animation and completed a guided tutorial level to ensure mechanical comprehension. To maintain experimental consistency and manage session duration, a 2-min maximum was enforced for this phase. While this approach resulted in slightly different levels of initial task exposure, the researcher provided a standardized verbal explanation of the game's control scheme and objectives to any participant who reached the 2-min limit without completion. This ensured that all participants possessed a baseline conceptual understanding of the mechanics before the eye-tracking data collection began. Before beginning the main experimental task, participants were provided with standardized verbal instructions to ensure consistent task framing across the sample. Specifically, participants were instructed to "navigate the level to collect all four keys and reach the exit gate as efficiently as possible." No explicit instructions were given regarding 'enjoyment' or 'speed-running' to avoid biasing the naturalistic distribution of gaze between the avatar and the environment, thereby allowing the internal moderators of game literacy and affective valence to emerge organically. When starting the experiment, the soundproof booth door was closed, while the researcher observed from an external monitor. Participants played until game completion (4 puzzles) or for a maximum of 15 min (average completion time was approximately 9 min)The session concluded with a post-game enjoyment survey.

2.4. Data availability statement

The game utilized in this study, *Vacillate*, is publicly available for review and replication at https://katesalembier.itch.io/vacillate. However, the anonymized eye-tracking dataset and participant survey responses are not available in a public repository due to ethical constraints. The institutional ethical approval and informed consent obtained for this research did not include provisions for the public deposition of raw or processed participant eye-tracking (biometric) data. Requests for further information regarding the analytical scripts or the research framework may be directed to the corresponding author.

2.5. Statistical analysis

To examine the relationship between game literacy, attentional distribution, and task performance, we employed a two-stage analytical framework. First, preliminary Pearson correlation analyses were conducted for descriptive purposes to examine bivariate linear relationships and to check for potential multicollinearity among predictors. Second, we conducted hierarchical regression analyses, which served as the sole confirmatory tests for our hypotheses. This approach ensured that hypothesis testing accounted for the shared variance between predictors (e.g., expertise and affect) and provided a more robust estimate of their unique contributions to the dependent variables.

Given the exploratory nature of this study and the multiple eye-tracking metrics under consideration, a Principal Component Analysis was first employed to reduce the dependent variable space and minimize the risk of Type I error.

3. Results

3.1. Preliminary analysis

Before testing our primary hypotheses, we assessed the data for completeness and attrition. Of the 52 participants who began the study, 52 successfully completed the entire game and post-game survey (100% retention). Seven participants were excluded from the final eye-tracking analysis due to calibration failure or data loss, resulting in a final N of 45. The training period was checked; 45 participants (83.3%) completed the training, while 7 participants (16.7%) did not. A t-test revealed no significant difference in task duration between the two groups ($p = .55$; Completed: $M = 527.64$s, $SD = 154.87$s; Not-completed: $M = 495.07$s, $SD = 122.94$s).

A post-hoc sensitivity analysis was conducted using G*Power 3.1 to determine the minimum effect size detectable with our final sample. For a multiple linear regression with 6 predictors and N = 45, at a conventional alpha of 0.05 and a power of 0.80, the study was powered to detect a large effect size of $f2 = 0.36$. While we acknowledge the study is underpowered to detect small-to-medium effects, our primary finding for Game Literacy ($R^2 = 0.334$; $f2 = 0.50$) exceeds this threshold, suggesting sufficient power for the identification of this large effect. Nonetheless, findings are presented as exploratory pending replication in a larger, more diverse sample.To improve the interpretability of regression coefficients, the game performance metric was converted from milliseconds to minutes. The categorical Gender variable was dummy coded in all regression models, using the 'woman' category as the reference group.

See Table 2 for descriptive statistics for all predictors and criterion

Table 2
Descriptive statistics for predictor and criterion variables (N = 45).

Variable	Mean	SD	Range (Min–Max)
Predictor Variables			
Age	24.22	7.09	17.00 – 51.00
Game Literacy	18.69	11.4	0.00 – 50.00
Enjoyment Rating	3.47	0.94	1.00 – 5.00
Global Gameplay Metrics			
Game Completion Time (s)	523.72	144.38	26.03 – 707.58
Global Eye-Tracking Metrics			
Dwell Time (%)	11.66	7.77	4.14 – 51.17
Fixation Count	190.89	220.14	7.00 – 1439.00
Revisit Count	122.84	63.84	9.00 – 309.00
Avg. Fixation Duration (ms)	224.89	35.12	155.81 – 320.44
Peak Saccade Velocity (°/s)	157.95	34.13	88.56 – 224.33
Avg. Saccade Duration (ms)	27.31	5.72	19.49 – 42.20
Fixation Dispersion	0.26	0.07	0.15 – 0.57

variables and Table 3 for the full correlation matrix.

3.2. Eye-tracking data quality

Eye-tracking data quality was assessed immediately following a 9-point calibration procedure. A minimum accuracy threshold of 0.5° of visual angle was established; participants failing to meet this threshold after two calibration attempts were excluded from the analysis (n = 3). Missing gaze segments shorter than 75 ms were accounted for using standard gap-fill interpolation within iMotions. Across the retained sample, the average gaze validity was 88.01%, indicating a data loss of approximately 11.99% due to blinks or momentary track loss.

The analysis was strictly confined to the main gameplay task. While calibration occurred prior to the training phase, the training puzzle was excluded from the processing pipeline. No additional adjustment period or temporal segmentation was applied to the start of the task. A check for outliers (defined as ±2.5 standard deviations from the mean) was performed on all eye-tracking variables, including saccade peak velocities (ranging from 122.48 to 160.01); however, no data points met the criteria for exclusion.

Preliminary distributional analysis confirmed that the proportion-based metrics (e.g., dwell time percentage) were centrally distributed ($M = 71.5\%$) and safely removed from the 0% and 100% boundaries.

3.3. Attentional allocation and monitoring (avatar area)

Descriptive statistics for all eye-tracking dependent variables, including means and standard deviations for dwell time percentage, fixation count, revisit count, peak saccade velocity, and fixation dispersion across the primary areas of interest, are summarized in Table 4.

As noted by the extremely high intercorrelations ($r > 0.93$) among dwell time percentage, fixation count, and revisit count, these metrics were collapsed into a single index (see Supplementary Materials for the full correlation matrix). To address the intercorrelation of eye-tracking metrics, a composite Avatar Monitoring Index was used as the dependent variable. A hierarchical regression was conducted to determine the relative influence of expertise and affect on monitoring behavior.

In Step 1, the control model (Gender and Game Literacy) accounted for 14.8% of the variance and approached significance, $F(2,37) = 3.21$, $p = .052$. Within this step, Game Literacy was a significant negative predictor ($\beta = -0.454$, $p = .017$).

In Step 2, the addition of Enjoyment significantly improved the model, $\Delta R^2 = 0.125$, $F(1,36) = 6.18$, $p = .018$. The final model was statistically significant, $F(3,36) = 4.50$, $p = .009$, $R^2 = 0.273$. In this final model, Game Literacy remained a robust negative predictor ($\beta = -0.636$, $p = .002$), indicating that higher expertise leads to reduced avatar monitoring. Conversely, Enjoyment emerged as a significant positive predictor ($\beta = 0.401$, $p = .018$), suggesting that higher game enjoyment is associated with increased attention toward the player avatar. The hierarchical regression summary is presented in Table 5.

3.4. Saccadic processing and cognitive efficiency (puzzle area)

To evaluate H2, we examined saccadic metrics within the puzzle AOI to determine how expertise and affect influence information processing. The analysis revealed that Game Literacy was the primary correlate of Peak Saccade Velocity, exhibiting a strong positive relationship, $r(43) = 0.77$, $p < .001$. This indicates that as players become more literate in the game's mechanics, their scanning speed increases significantly. Enjoyment showed a small negative correlation with peak velocity, $r(43) = -0.20$, $p = .196$, but did not reach statistical significance at the bivariate level.To analyze the average saccade direction, a variable transformation was required, as the circular nature of angular data (0-360°) violates the core assumptions of standard linear regression. Therefore, the raw directional angle for each participant was converted into a linear variable, termed the "Downward Saccade Component," to facilitate the analysis. This was achieved by calculating the sine of each participant's average saccade angle (after converting it from degrees to radians). The resulting component maps the circular data onto a continuous linear scale ranging from −1.0 to +1.0, where a value of +1.0 represents a purely downward average saccade direction (equivalent to 90°), a value of −1.0 represents a purely upward direction (270°), and a value of 0.0 represents a purely horizontal direction (0° or 180°). This transformed variable was used as the dependent variable in a hierarchical regression. However, the model was non-significant, $F(3,36) = 0.83$, $p = .567$, indicating that neither expertise nor enjoyment significantly predicted a consistent vertical bias in saccade direction beyond the structural requirements of the interface.To evaluate the unique contribution of enjoyment to saccade dynamics, we employed a hierarchical regression predicting Peak Saccade Velocity (Puzzle Area), a marker of scanning efficiency and cognitive friction. To resolve the shared variance between expertise and affect, all continuous predictors were mean-centered.

In Step 1, the model (Gender and Game Literacy) was significant, $F(2,42) = 32.37$, $p < .001$. Within this model, Game Literacy emerged as a powerful positive predictor ($B = 0.202$, $\beta = 0.794$, $p < .001$, 95% CI [0.15,0.25]), indicating that as expertise increases, participants exhibit significantly faster saccadic transitions when scanning puzzle elements.

In Step 2, the addition of Enjoyment did not result in a significant increase in variance explained, $\Delta R^2 = 0.004$, $F(1,41) = 0.46$, $p = .500$. The final model remained significant, $F(3,41) = 21.46$, $p < .001$, $R^2 = 0.611$. In this final model, Game Literacy remained the sole significant predictor ($\beta = 0.780$, $p < .001$), while Enjoyment was non-significant ($\beta = -0.070$, $p = .500$). The hierarchical regression summary is presented in Table 6.

4. Global gaze dynamics and ergonomic pattern to playing the game (screen area)

The distribution of the average saccade direction for all players is visualized in Fig. 6.

Note. The circular distribution of average saccade directions is visualized in Fig. 4 using a rose diagram. The analysis reveals a highly concentrated unimodal distribution, with a prominent peak centered at the 180° horizontal axis. This indicates a dominant preference for lateral scanning patterns during puzzle engagement, with the vast majority of saccades aligning with the horizontal plane. The circular mean of the distribution was calculated to be 184.3°, confirming that gaze transitions were predominantly horizontal, reflecting the linear spatial arrangement of the task-relevant information.

Table 3
Descriptive statistics and pearson correlations for primary study variables.

Variable	M	SD	1	2	3	4	5
1. Gender	0.7	0.46	—				
2. Game Literacy	17.48	10.19	−0.55**	—			
3. Enjoyment	3.5	0.99	−0.28	0.47**	—		
4. Game Completion Time (s)	531.46	145.83	0.22	0.12	−0.14	—	
5. Avatar Monitoring Index	−0.04	1.4	−0.06	−0.28	0.19	−0.72**	—
6. Peak Saccade Velocity	157.26	34.18	0.33*	−0.22	−0.43**	0.1	−0.17

Note. $N = 45$. *$p < .05$,$p < .01$.

Table 4
Descriptive statistics for eye-tracking metrics across primary areas of interest.

Variable	Screen (Global)	Character	Top (Avatar Area)	Bottom (Puzzle Area)
Dwell Time (%)	75.65 (11.61)	54.59 (10.91)	23.47 (9.27)	7.45 (4.43)
Fixation Count	1422.84 (421.61)	900.92 (288.38)	497.37 (209.23)	171.16 (107.84)
Revisit Count	—	223.21 (67.64)	239.34 (78.15)	84.76 (38.70)
Peak Velocity (°/s)	154.73 (28.28)	102.72 (15.81)	118.61 (37.78)	160.60 (32.10)
Fixation Dispersion	0.32 (0.06)	0.33 (0.06)	0.38 (0.09)	0.25 (0.05)

Note. Values are presented as Mean (Standard Deviation). The Screen (Global) metric represents the total valid tracking ratio across the task. Dwell time percentages do not sum to the Screen total because the Areas of Interest (AOIs) are not mutually exclusive. Specifically, the Character AOI is a central anchor that spatially overlaps with the Top and Bottom screen halves; thus, a single fixation on the character may be registered across multiple AOIs simultaneously.

Table 5
Hierarchical regression analysis predicting the avatar monitoring index.

Predictor	B	SE	β	t	p	95% CI [LL,UL]
Step 1						
Constant	1.713	0.753		2.275	0.029	[0.18,3.24]
Gender	−0.941	0.549	−0.311	−1.715	0.095	[−2.05,0.17]
Game Literacy	−0.062	0.025	−0.454	−2.5	0.017*	[−0.11,−0.01]
Step 2						
Constant	0.131	0.95		0.138	0.891	[−1.80,2.06]
Gender	−0.904	0.514	−0.299	−1.757	0.087	[−1.95,0.14]
Game Literacy	−0.087	0.025	−0.636	−3.434	0.002	[−0.14,−0.04]
Enjoyment	0.569	0.229	0.401	2.486	0.018*	[0.11,1.03]

Note. CI = confidence interval; LL = lower limit, UL = upper limit.
*$p < .05$.

Table 6
Hierarchical regression results for global peak saccade velocity.

Predictor	B	SE	β	t	p	95% CI [LL,UL]
Step 1						
Constant	20.451	42.165		0.485	0.63	[−64.64,105.54]
Gender	7.693	8.884	0.086	0.866	0.391	[−10.24,25.62]
Game Literacy	0.202	0.025	0.794	7.987	<0.001*	[0.15,0.25]
Step 2						
Constant	61.038	73.252		0.833	0.41	[−86.90,208.97]
Gender	5.971	9.293	0.067	0.642	0.524	[−12.80,24.74]
Game Literacy	0.199	0.026	0.78	7.628	<0.001*	[0.15,0.25]
Enjoyment	−0.302	0.444	−0.07	−0.68	0.5	[−1.20,0.60]

Note. N = 45. CI = confidence interval; LL = lower limit, UL = upper limit. For Step 1: R2 = 0.607, F(2,42) = 32.37, $p < .001$. For Step 2: $\Delta R^2 = 0.004$, F(1,41) = 0.46, $p = .500$.
*$p < .001$.

4.1. Attentional precision and visual dispersion

Finally, to test H3, which predicted that higher game literacy would lead to more compressed and strategically focused gaze patterns, a multiple linear regression was conducted to predict the global dispersion of fixations. The model included player age, gender, game literacy, enjoyment, and game duration as predictors. The overall regression model was statistically significant, $F(4,40) = 4.12$, $p = .007$, and accounted for 29.1% of the variance in global fixation dispersion. Consistent with H3, an examination of the individual predictors revealed that Game Literacy was a statistically significant negative predictor of fixation dispersion ($B = -0.005$, SE = 0.002, $p = .012$), indicating that more experienced players exhibited more spatially concentrated gaze patterns. No other predictors were found to be statistically significant. See Fig. 7 for a representaion of scan path patterns.

4.2. Diagnostic testing

Prior to interpreting the regression coefficients, we conducted a comprehensive assessment of the standard assumptions for multiple linear regression. For our primary model (Global Peak Saccade Velocity), the residuals were found to be normally distributed (Shapiro-Wilk $W = 0.98$, $p = .377$) and exhibited homoscedasticity (Breusch-Pagan $p = .361$). Multicollinearity was minimal; the Variance Inflation Factors (VIF) for all predictors, including the mean-centered literacy and enjoyment scores, were below 1.65, well within the acceptable range. Finally, an analysis of Cook's Distance (Max D = 0.24) and leverage statistics confirmed the absence of influential outliers that might have distorted the reported effects.

4.3. Predictors of performance and player experience

A specific analysis was conducted to test the hypothesis that game performance, as measured by total completion time, was related to player enjoyment. A Pearson correlation revealed a weak, non-significant negative relationship between game duration and enjoyment, $r(43) = -0.13$, $p = .384$. To confirm this, a linear regression was performed to predict enjoyment from game duration. The overall model was not statistically significant, $F(1, 43) = 0.77$, $p = .384$, and accounted for a negligible 1.8% of the variance in enjoyment ($R^2 = 0.018$). Thus, game performance was not found to be a significant predictor of player enjoyment.

5. Summary of results

The results across all three analyses provide consistent support for

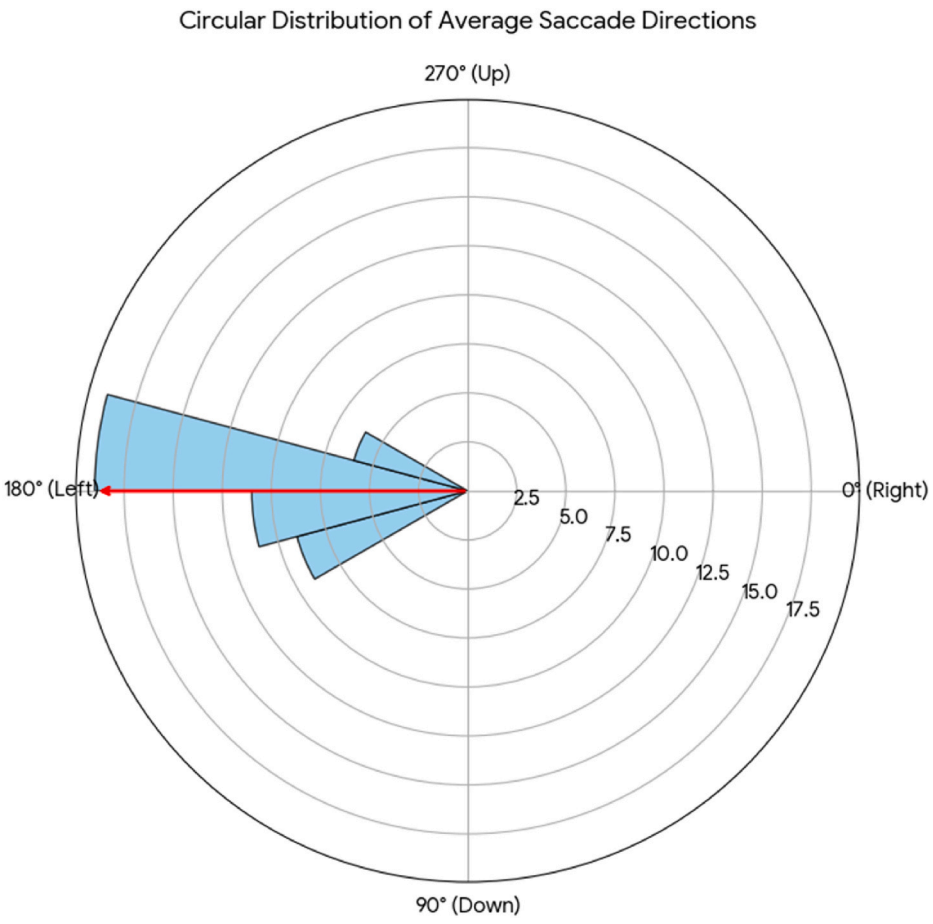


Fig. 6. Average saccade direction for all players.

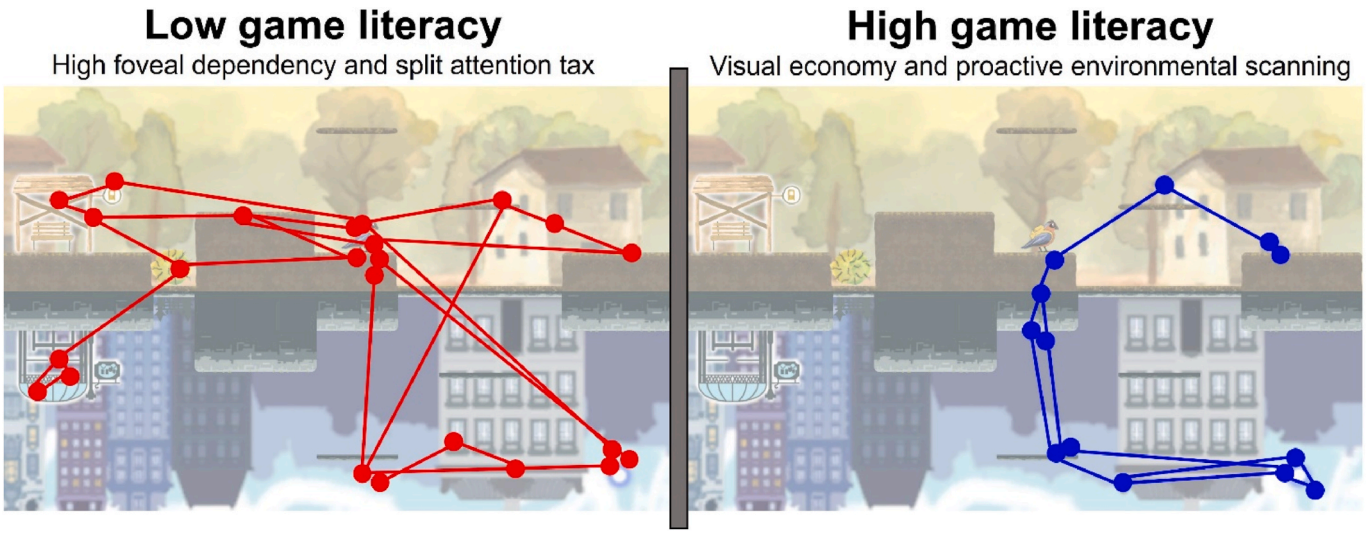


Fig. 7. Representative 10-Second Gaze Scanpaths Contrasting Visual Strategies

Note. Comparison of typical gaze patterns for players with low game literacy (red) and high game literacy (blue). The left panel (low game literacy) illustrates high foveal dependency and the split-attention tax through frequent, high-amplitude vertical transitions between the bird avatar and the puzzle area. The right panel (high game literacy) demonstrates visual economy and proactive environmental scanning, with gaze concentrated on upcoming platforms and critical puzzle information rather than the avatar. These scanpaths serve as representative illustrative examples to provide a qualitative visual shorthand for the predictive relationships identified in the regression models.

the predictive power of Game Literacy within our proposed framework. While lower game literacy was characterized by a reactive foveal dependency on the player avatar, increased expertise facilitated a shift toward proactive environmental scanning and higher peak saccade velocities, with the latter serving as an objective marker of motor automaticity. Interestingly, while Enjoyment remains a critical component of

the play experience, it did not significantly predict the efficiency of these oculomotor markers in the same capacity as literacy. Together, these findings suggest that the transition from novice to expert is defined by a fundamental restructuring of visual strategy, moving from an attentional tax on character control to a streamlined, predictive engagement with the game world.

6. Discussion

6.1. The push and pull of attentional allocation

The analysis of the composite Avatar Monitoring Index provides strong evidence for H1. By collapsing highly correlated eye-tracking metrics, we identified a unified latent construct of monitoring intensity. Game Literacy emerged as the most powerful negative predictor of this index, confirming that expertise allows players to decouple their visual attention from the avatar to focus on the wider environment. However, the hierarchical model revealed a compelling 'push-pull' dynamic regarding attentional allocation. While Literacy 'pushes' the gaze away from the avatar to facilitate spatial efficiency, Enjoyment acts as an affective pull. This suggests that players who find the experience more rewarding may be more likely to re-invest visual resources back into the player character. Additionally, avatar monitoring is not solely a tax paid by the player with less experience, but also a reward sought by the engaged player, perhaps to appreciate character animations or feedback once the cognitive load of control has been automated. Regarding H2, the results provide a more nuanced look at oculomotor efficiency. While we did not find the expected relationship with Enjoyment, we found that Game Literacy was the primary and positive predictor of peak velocity. This suggests that in the context of a cognitively demanding task like puzzle-solving, high-velocity saccades are a hallmark of expert scanning. Experienced players process the visual grammar of the game more rapidly, allowing for higher information throughput and faster spatial transitions than their less-literate counterparts.

Our findings related to the ergonomic pattern of the game involved a dominant left-to-right visual flow. While all players adhered to this general pattern, those with higher game literacy navigated the environment with a more spatially concentrated gaze. This reduction in global fixation dispersion suggests a more refined spatial economy, consistent with the interpretation that expertise allows for more focused task engagement. In contrast, players with lower literacy exhibited a more dispersed gaze pattern, suggesting a more frantic, exploratory scanning strategy. Together with the AOI results, this indicates that the more expert gaze is not merely more focused, but is specifically released from the foveal dependency of the avatar, allowing for a more concentrated deployment of attention within the task-relevant puzzle area.

6.2. Defining the pathways of interactive visual economy

Our results support a multi-dimensional model of interactive visual economy, which we define as the optimized allocation of gaze resources to maximize task-relevant information extraction while minimizing cognitive and oculomotor friction. In this framework, visual economy is not a single variable, but a state achieved through three distinct pathways: (1) Attentional Economy, where Game Literacy allows players to decouple their gaze from the avatar (reduced Avatar Monitoring); (2) Temporal Economy, where information is processed at higher speeds (increased Peak Velocity); and (3) Spatial Economy, where gaze is concentrated within critical task areas (reduced Fixation Dispersion).

7. The core attentional conflict: attentional tax to foveal dependency

The game's design creates a fundamental split-attention conflict. Players are forced to constantly shift their gaze and cognitive resources between two demanding, spatially separated tasks: motor control and navigation (platforming on top) and information extraction and problem-solving (puzzles on the bottom). Our analysis reveals that a player's ability to efficiently manage this conflict is the primary predictor of their strategic success. The large strength of this relationship suggests that the influence of game literacy is not merely statistically significant, but represents a practically meaningful driver of how players navigate the fundamental conflicts of interactive attention.

Interestingly, while the data from the avatar AOI shows that less experienced players pay a "cost" in the form of avatar monitoring, these results suggest a necessary post-hoc refinement of our initial 'attentional tax' framework. Specifically, the data indicate that the term "tax" may be insufficient. Less experienced players have not yet automated the feel of the platforming controls, requiring them to dedicate significant visual attention to confirming their avatar's position and action. Every moment spent looking at the avatar is a moment *not* spent gathering information from the puzzle area below. However, it is not the case that decreasing avatar monitoring would inherently improve performance; instead, it is an allocation strategy. For inexperienced players, gaze serves as a primary feedback loop for motor coordination as they learn the game. This proposed construct of foveal dependency acts as a bridge for the motor system that has yet to adjust to the feel of the game, or how it controls.

Our findings suggest two distinct ways a player might struggle, each with its own unique eye-tracking signature. First, more novice players, defined by low Game Literacy, show gaze behavior that is "stuck" on the top half of the screen. They exhibit high dwell time, fixation counts, and avatar revisit rates. They are unable to effectively extract information from the puzzle area because their attention is almost entirely consumed by the platforming sub-task. The inexperienced player resolves the attentional conflict by largely ignoring the puzzle task. They are trapped in an avatar monitoring loop on the top screen because basic survival (e.g., not colliding with obstacles) is their most pressing concern. They cannot afford to make the long, vertical saccades down to the puzzle area because they lack the confidence (or automation) to let their avatar operate unwatched in their peripheral vision; they are stuck in foveal dependency.

Second, even when a player has successfully transitioned their gaze to the puzzle area, the quality of their scanning remains a function of their underlying expertise. Our analysis reveals that high-velocity saccades, a marker of expert information throughput, are predicted exclusively by Game Literacy. This suggests that the frantic scanning often observed in struggling players may not be a byproduct of their emotional state, but rather a mechanical limitation of their lower game literacy. In this view, the expert knows where to look in addition to possessing the oculomotor efficiency to process that information at a higher temporal frequency.This distinction is critical: low game literacy imposes a dual burden on the player,who may struggle because they don't know where to look (they're stuck on the avatar, remaining stuck in foveal dependency), and because they don't know how to look (their search is inefficient, lacking the saccadic velocity required)).

8. The affective pull: how enjoyment modulates gaze

Our results suggest that enjoyment influences eye movements through a specific affective pull rather than by altering the physical speed of information processing. Enjoyment does not dictate the quality or velocity of scanning when solving puzzles. Instead, saccadic velocity is driven almost exclusively by Game Literacy. This suggests that the deliberateness of a player's scan is a function of their internalized skill rather than their emotional state. The primary role of enjoyment is found in the priority of attention, where it acts as a significant positive predictor of avatar monitoring. While expertise pushes the gaze away from the avatar for the sake of efficiency, higher enjoyment pulls the fovea back toward it. This indicates that a high-enjoyment state is not defined by calm scanning, but by attentional agency. The player utilizes their game literacy to process puzzle information at high-velocity expert

speeds, which in turn grants them the attentional surplus to reinvest their gaze into the character they are controlling.

9. Applying the dual-component framework

Synthesizing these findings, our proposed a dual-component framework of interactive attention extends Cognitive Load Theory into the oculomotor domain. This model offers a falsifiable account of the novice-to-expert transition, characterized by two mechanistically dissociated components attentional efficiency (related to expertise) and attentional state (related to the interaction of expertise and enjoyment).

The first component is that expertise is related to attentional allocation. The avatar AOI analysis shows that expertise (high Game Literacy) manifests as the ability to stop looking at the avatar. This is the foundational step towards skilled play. The inexperienced player, trapped in a feedback loop of avatar monitoring, is fundamentally playing a different, more constrained game. The second component is that expertise and enjoyment have a push-pull relationship with the priority of monitoring. While literacy pushes attention away from the avatar for the sake of efficiency, enjoyment pulls it back for the sake of engagement.

From this, we propose a model of player progression. First, the data are consistent with a model where a player achieves attentional efficiency through literacy, freeing the fovea from its dependency on the avatar and increasing visual throughput in the task area. Once this efficiency is achieved, the player enters an attentional state of optionality. In this state, gaze is no longer a dependent on motor control; instead, it can be re-allocated based on the player's affective state, either toward rapid puzzle processing or back toward the avatar for aesthetic and emotional engagement.

9.1. Disentagling expertise and affect

By assessing expertise and affect within a single hierarchical model, we are able to demonstrate that these constructs do not merely overlap, but instead map onto dissociable oculomotor pathways: one governing the physical tempo of information throughput and the other governing the priority of attentional allocation.

Because Game Literacy and Enjoyment were significantly correlated, it was necessary to ensure that our regression results were not distorted by multi-collinearity. VIF confirmed that the constructs were sufficiently distinct to allow for the assessment of their unique contributions. Our findings suggest that while these variables share variance, they are associated with different pathways. Expertise serves as oculomotor behavior, dictating the physical speed of information extraction (peak velocity). Affect serves more as a modulator, dictating how attentional resources are allocated once task demands are met (avatar monitoring). We suggest that this indicates that the observed oculomotor signatures of an expert are not merely a byproduct of being more satisfied with the game, but represent a distinct cognitive adaptation to interactive environments.

It is notable that while Game Literacy served as a robust predictor of these oculomotor signatures, raw task performance (completion time) did not. While all participants successfully completed the task, the lack of a correlation suggests that cumulative expertise is a more sensitive indicator of a player's visual grammar than simple task speed. This indicates that the observed oculomotor patterns of an expert represent a more fundamental shift in how they prioritize and process interactive information.

10. Study limitations

It is important to acknowledge that the present study should be considered exploratory in nature (was not pre-registered). While the research was guided by established theoretical frameworks concerning split-attention and cognitive load, the findings presented here are intended to be hypothesis-generating rather than confirmatory. Specifically, a post-hoc sensitivity analysis indicates the study was powered to detect large effects, such as the one observed for Game Literacy, but remains underpowered for detecting medium-sized relationships. Future research utilizing a pre-registered, confirmatory design is necessary to replicate these results and further validate the specific eye-tracking markers identified as predictors of interactive expertise.

One limitation to note in the current study was the skewed gender in our sample. The demographic was predominantly female-identifying (63.5%), with significantly lower representation of male (26.9%), nonbinary (7.7%), and apagender (1.9%) participants from the United States. While our predominantly female, university-aged sample may limit generalizability to the broader population, the inclusion of nonbinary and apagender participants represents a significant representational strength that addresses the historical male-bias in gaming research. Future studies should examine a balanced cross-section of demographics to determine if the transition from foveal dependency to motor automaticity remains consistent across different demographic groups.

Furthermore, the use of a single custom-built 2D paradigm and a single-item enjoyment measure, while necessary for experimental control and ecological validity, represents a constraint on the generalizability and psychometric depth of the findings. Future studies should examine a balanced cross-section of demographics and utilize validated multi-item scales (e.g., the GEQ or IMI) to determine if the transition from foveal dependency to motor automaticity remains consistent across different groups and genres.

Finally, the present study utilized session-level aggregation for all eye-tracking metrics. While this provided a stable measure of between-participant differences, it may have obscured the temporal microdynamics of within-session learning. Additionally, our spatial AOI definitions likely captured a broader strategy of character-proximate scanning rather than exclusive avatar-fixation. Consequently, our proposed developmental model serves as a theoretical extrapolation from cross-sectional data rather than a direct observation of longitudinal transitions. Future work should utilize multilevel or mixed-effects modeling to examine these within-participant shifts over time, across gameplay, identifying the specific moments when a lower game literacy player's gaze begins to mirror the high-velocity, low-monitoring patterns characteristic of the player with higher game literacy.

11. Future directions

Eye-tracking signatures provide an objective, real-time window into attentional strategy that can directly inform the design of adaptive, state-aware systems. While exploratory, these findings suggest several hypotheses that future design research could exploresuch as creating tutorials that might train attention away from the avatar to more rapidly induce a state of enjoyable flow. Future design research could explore tutorials that, instead of teaching controls, focus on teaching visual strategy. Based on the current results, one could hypothesize an effective tutorial that might train players away from avatar fixation. For instance, as using a short “training level” where level design builds the players’ trust in their peripheral vision and learn to dedicate their gaze to the puzzle area without fearing for their avatar.With the rise in accesible consumer eye-trackers, future studies should explore the possibility of using eye-tracking metrics themselves to help players learn in real-time. For example, a system that detects a sustained state of very high saccade velocity in the puzzle area could infer that a player is overwhelmed. In response, the game could trigger a subtle hint or simplify a minor puzzle element to guide the player out of a frustrating loop and back into a state of enjoyable flow.

While we initially hypothesized that inexperienced players might have an attentional tax compared to more experienced players, the findings indicate a state of foveal dependency is more appropriate. Nonetheless, future research should investigate the actual functional

necessity of such gaze behavior. Specifically, at what point does avatar monitoring transition from a required feedback loop into a redundant habit? By isolating the moment of such a proprioceptive automaticity, where the player begins to "feel" the jump without seeing it, designers can better time the introduction of complex, spatially separated puzzle information Beyond motor automaticity, foveal dependency may also be influenced by unmeasured factors like spatial working memory or task-based anxiety. While our model disentangles affective interest (enjoyment) from literacy, future research incorporating psychometric assessments is needed to isolate these alternative cognitive and emotional drivers of attentional allocation.

11.1. Conclusions

The use of a custom-built experimental paradigm allowed us to isolate the fundamental cognitive mechanics of split-attention within a 2D environment, effectively removing the noise inherent in commercial game features. By contrasting the engaging, naturalistic experience of Vacillate with the contrived tasks typical of traditional laboratory attention research, we provide a more ecologically valid account of how cognitive load and affect interact in modern digital environments. Consequently, the transition from foveal dependency to attentional agency observed here may serve as a baseline for other interactive environments, such as dashboard design or remote piloting, where users must navigate the same tension between motor control and information extraction. Rather than being limited to a single title, these findings suggest that the visual grammar of expertise is a measurable, generalizable property of skilled human-computer interaction.

The transition to expertise is a shift in visual agency. By quantifying attentional tax as a measurable foveal dependency, we show that expertise breaks the gaze's bond to the avatar. However, our model uniquely shows that enjoyment can re-establish this bond by choice. To facilitate a positive attentional state, designers could consider scaffolding the eyes away from the avatar for efficiency, while imbuing the avatar with enough affective value to make looking back a rewarding experience.

CRediT authorship contribution statement

Joshua Juvrud: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Katherine Salembier:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Conceptualization. **Margot Delany:** Writing – review & editing, Writing – original draft, Investigation. **Shiloh Frost:** Writing – review & editing, Writing – original draft, Investigation. **Xingcen (Lucy) Liu:** Writing – review & editing, Writing – original draft, Investigation. **Adrian Łoboda:** Writing – review & editing, Writing – original draft, Software. **Evangelia Sotiriou:** Writing – review & editing, Writing – original draft, Software.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chb.2026.109055.

Data availability

Data will be made available on request.

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